

Chapter 1 — Top 15 Exam-Ready Questions & Answers

Q1. What is Software Project Management (SPM), and is it fundamentally different from ordinary project management?

বাংলা অর্থ

Software Project Management (SPM) হলো software project-কে **plan, organize, monitor** এবং **control** করা, যাতে নির্দিষ্ট সময়, খরচ, scope এবং quality-এর মধ্যে project শেষ করা যায়।

SPM আর সাধারণ Project Management মূলত একই ধরনের management principles follow করে। পার্থক্য হলো **software-এর nature আলাদা**, তাই software project-এ কিছু বিশেষ সমস্যা থাকে।

Exam-Ready Answer

Software Project Management is the process of planning, organizing, monitoring, and controlling a software project to achieve its objectives within the specified time, cost, scope, and quality.

It is not fundamentally different from ordinary project management, as both involve planning, resource management, scheduling, cost control, risk management, and quality control.

□ মনে রাখো

SPM = Plan + Organize + Monitor + Control

Q2. How does Longman's dictionary define a "project," and what are its two key distinguishing factors?

বাংলা অর্থ

Longman's dictionary অনুযায়ী project হলো:

- একটি **specific plan/design**
- একটি **planned undertaking**
- একটি বড় ধরনের undertaking

Project-এর দুটি গুরুত্বপূর্ণ বৈশিষ্ট্য:

1. **Planning** থাকতে হবে

2. কাজটি significant/বড় ধরনের হতে হবে

Exam-Ready Answer

According to Longman's dictionary, a project is a specific plan or design, a planned undertaking, or a large undertaking.

The two key characteristics of a project are:

1. It requires **planning**.
2. It involves a **significant or substantial task**.

☐ Shortcut

Project = Planned + Significant

Q3. Differentiate between "Jobs," "Exploration," and "Projects."

বাংলা অর্থ

এখানে **uncertainty** বা অনিশ্চয়তার ভিত্তিতে তিনটি বিষয় আলাদা করা হয়েছে।

Job

একই ধরনের কাজ বারবার করা হয় এবং **uncertainty** খুব কম।

Example: প্রতিদিন employee attendance process করা।

Exploration

কী result হবে সেটাই নিশ্চিত নয়।

Example: নতুন কোনো medicine আবিষ্কার করা।

Project

কী achieve করতে হবে জানা আছে, কিন্তু development-এর মধ্যে কিছু **uncertainty** থাকে।

Example: একটি Banking Software তৈরি করা।

Exam-Ready Answer

Jobs involve repetitive, well-defined, and well-understood tasks with very little uncertainty.

Exploration involves highly uncertain outcomes where the final result may not be known in advance.

Projects lie between jobs and exploration. They have specific targets but involve a moderate level of uncertainty and risk.

□ মনে রাখো

Job → **Low uncertainty**

Project → **Medium uncertainty**

Exploration → **High uncertainty**

Q4. What are the core characteristics that make a task "project-like"? বাংলা অর্থ

কোনো কাজকে Project বলা যাবে যখন সেটি:

- Routine কাজ নয়
- আগে থেকে planned
- Specific target আছে
- Customer-এর জন্য করা হয়
- বিভিন্ন ধরনের specialist লাগে
- কয়েকটি phase থাকে
- নির্দিষ্ট time ও resource-এর মধ্যে করতে হয়
- কাজটি বড়/complex হতে পারে

Exam-Ready Answer

A task is considered project-like if it is non-routine, planned, and aimed at a specific target.

It usually involves work for a customer, requires several specialized skills, and consists of several phases.

It is also constrained by time and resources and is generally large or complex.

□ Shortcut

Non-routine + Planned + Target + Customer + Time + Resources

Q5. Why are software projects considered more problematic to build and manage than other engineered artifacts?

বাংলা অর্থ

Software project-এর ৪টি বড় সমস্যা আছে। এটাকে ICCF দিয়ে মনে রাখা যায়।

I — Invisibility

Software-এর physical shape নেই। তাই project কতদূর এগিয়েছে সেটা চোখে দেখা কঠিন।

C — Complexity

Software-এর অনেক component এবং interaction থাকে।

C — Conformity

Software-কে existing system, rules, standards এবং user requirements-এর সাথে মিল রেখে চলতে হয়।

F — Flexibility

Software সহজে পরিবর্তন করা যায়। তাই customer বারবার requirement change করতে চায়।

Exam-Ready Answer

Software projects are more difficult to manage because of four major characteristics:

Invisibility: Software has no physical form, making progress difficult to see and measure.

Complexity: Software systems contain many components and interactions, making them difficult to understand and maintain.

Conformity: Software must conform to existing systems, standards, rules, and user requirements.

Flexibility: Software is relatively easy to change, which can lead to frequent requirement changes and increased cost and development time.

★ Exam Shortcut

ICCF = Invisibility + Complexity + Conformity + Flexibility

Q6. What are the three primary activities covered by project management?

বাংলা অর্থ

Project Management-এর তিনটি প্রধান activity:

1. Feasibility Study

Project করা সম্ভব এবং business-এর জন্য worthwhile কিনা দেখা।

2. Planning

Project feasible হলে কীভাবে project করা হবে তার plan তৈরি করা।

3. Execution

Plan অনুযায়ী actual কাজ করা।

Exam-Ready Answer

The three primary activities of project management are:

1. **Feasibility Study:** Determining whether the project is technically feasible and worthwhile from a business perspective.
2. **Planning:** Preparing the plan for carrying out the project after it is considered feasible.
3. **Execution:** Implementing the plan while making necessary changes as the project progresses.

☐ Shortcut

Feasibility → Planning → Execution

Q7. Describe the first three phases of the ISO 12207 Software Development Life-cycle.

বাংলা অর্থ

ISO 12207-এর প্রথম তিনটি phase:

1. Requirements Analysis

Customer কী চায় সেটা বের করা।

2. Architecture Design

System-এর overall structure/components ঠিক করা।

3. Code and Test

Component-এর code লেখা এবং test করা।

Exam-Ready Answer

The first three phases of the ISO 12207 software development life-cycle are:

1. **Requirements Analysis:** Identifying customer requirements and converting them into requirements understandable by developers, including functional, quality, and resource requirements.
2. **Architecture Design:** Defining the major components of the system, including hardware, software, and organizational components.
3. **Code and Test:** Developing and testing the individual software components.

☐ Shortcut

Requirement → Design → Code/Test

Q8. Outline the final four phases of the ISO 12207 Software Development Life-cycle.

বাংলা অর্থ

প্রথমে individual component বানানো হলো। এখন সেগুলোকে একসাথে করে পুরো system তৈরি করতে হবে।

1. Integration

সব component একসাথে করা।

2. Qualification Testing

পুরো system test করা।

3. Installation

Customer environment-এ system install করা এবং users-কে training দেওয়া।

4. Acceptance Support

System চালু হওয়ার পর maintenance এবং enhancement support দেওয়া।

Exam-Ready Answer

The final four phases are:

1. **Integration:** Combining the individual components into a complete system.
2. **Qualification Testing:** Testing the complete system rather than individual software components.
3. **Installation:** Making the system operational, setting data and parameters, and providing user training.
4. **Acceptance Support:** Providing ongoing maintenance and enhancement after system delivery.

★ ISO 12207 Full Sequence

Requirements Analysis → Architecture Design → Code & Test → Integration → Qualification Testing → Installation → Acceptance Support

এটা exam-এর জন্য মুখস্থ করে রাখো।

Q9. Why is it necessary to categorize projects, and what are the main categories?

বাংলা অর্থ

সব project একইভাবে manage করা যায় না।

যেমন:

Banking Software আর **Car-এর embedded software** একইভাবে manage করা উচিত নয়।

তাই project-এর ধরন অনুযায়ী management approach পরিবর্তন করতে হয়।

প্রধান categories:

1. Information System

Business data/process manage করে।

2. Embedded System

Hardware-এর ভিতরে থাকা software।

আরেকভাবে:

Objective-based → Goal achieve করা

Product-based → Specific product তৈরি করা

Exam-Ready Answer

It is necessary to categorize projects because different types of projects require different management approaches and techniques.

Projects may be categorized as:

- **Information systems:** Systems mainly used to manage business information and data.
- **Embedded systems:** Software embedded within hardware devices.
- **Objective-based projects:** Projects aimed at achieving a particular objective.
- **Product-based projects:** Projects focused on developing a specific product.

Q10. What are the core activities involved in "management"?

বাংলা অর্থ

একজন Manager-এর প্রধান কাজগুলো হলো:

1. **Planning** → কী করতে হবে ঠিক করা
2. **Organizing** → কাজের arrangement করা
3. **Staffing** → লোক নির্বাচন করা
4. **Directing** → নির্দেশ দেওয়া
5. **Monitoring** → progress দেখা
6. **Controlling** → সমস্যা হলে ব্যবস্থা নেওয়া
7. **Innovating** → নতুন solution বের করা
8. **Representing** → stakeholder-এর সাথে যোগাযোগ করা

Exam-Ready Answer

The core activities of management include:

- **Planning:** Deciding what should be done.
- **Organizing:** Making the necessary arrangements.
- **Staffing:** Selecting appropriate people.
- **Directing:** Providing instructions and guidance.
- **Monitoring:** Checking progress.

- **Controlling:** Taking corrective actions when problems occur.
- **Innovating:** Developing solutions to new problems.
- **Representing:** Communicating and liaising with stakeholders.

☐ **Shortcut**

Plan → Organize → Staff → Direct → Monitor → Control → Innovate → Represent

Q11. What is the role of a Project Authority in setting objectives?

বাংলা অর্থ

Project Authority হলো project-এর senior decision-making authority।

তাদের কাজ:

- Project-এর **scope** নির্ধারণ করা
- Project-এর **cost/budget approve বা allocate** করা

এটি একজন ব্যক্তি অথবা একটি group হতে পারে।

যেমন:

- Project Board
- Steering Committee

Exam-Ready Answer

The Project Authority is responsible for defining the project scope and allocating or approving the project costs.

It may consist of a single person or a group such as a Project Board, Project Management Board, or Steering Committee.

☐ **Shortcut**

Project Authority = Scope + Cost

Q12. What does the acronym SMART stand for when setting project objectives?

বাংলা অর্থ

SMART objective মানে objective এমন হতে হবে যাতে সেটা clear এবং measurable হয়।

S — Specific

নির্দিষ্ট হতে হবে।

M — Measurable

মাপা যাবে।

A — Achievable

বাস্তবে achieve করা সম্ভব হতে হবে।

R — Relevant

Project-এর মূল উদ্দেশ্যের সাথে সম্পর্কিত হতে হবে।

T — Time-constrained

নির্দিষ্ট সময়ের মধ্যে achieve করতে হবে।

Exam-Ready Answer

SMART stands for:

S – Specific: The objective should be clear and well-defined.

M – Measurable: Achievement should be objectively measurable.

A – Achievable: The objective should be realistically achievable.

R – Relevant: The objective should be relevant to the purpose of the project.

T – Time-constrained: The objective should have a defined deadline.

★ Shortcut

Specific → Measurable → Achievable → Relevant → Time-constrained

Q13. How do Goals/Sub-objectives relate to Measures of Effectiveness?

বাংলা অর্থ

Main Goal হলো final target।

Sub-objectives হলো সেই target অর্জনের ছোট ছোট steps।

Measure of Effectiveness হলো objective আসলেই achieved হয়েছে কিনা সেটা মাপার পদ্ধতি।

Example

Goal:

Improve customer service.

Sub-objective:

Reduce customer complaints.

Measure:

Number of complaints per month.

Exam-Ready Answer

Goals or sub-objectives are smaller targets or steps that must be achieved to accomplish the main project objective.

Measures of effectiveness are used to determine objectively whether a goal or sub-objective has been achieved.

For example, customer satisfaction can be measured by the number of complaints received.

☐ **Shortcut**

Goal = কী achieve করব

Measure = Achieve হয়েছে কিনা কীভাবে জানব

Q14. Explain the concept of a "Business Case" in project management.

বাংলা অর্থ

Business Case হলো project করলে benefit বেশি হবে নাকি cost বেশি হবে—এটা যাচাই করা।

Cost:

- Development cost
- Operational cost

Benefit:

- Financial benefit
- Improved reputation
- Better customer service ইত্যাদি

যদি **Benefit** > **Cost**, তাহলে project worthwhile হতে পারে।

Exam-Ready Answer

A business case evaluates whether the benefits of a project are greater than its costs.

Costs include development and operational expenses, while benefits may be either quantifiable or non-quantifiable.

A project is considered worthwhile when its expected benefits justify or exceed its costs.

☐ **Shortcut**

Business Case = Benefit vs Cost

Q15. Describe the Management Control process with its sequential steps.

বাংলা অর্থ

Management Control-এর ধাপগুলো হলো:

1. Data

Raw information collect করা।

Example: 6,000 documents processed.

2. Information

Data-কে meaningful করা।

Example: প্রতিদিন 100 documents process হচ্ছে।

3. Comparison

Actual performance-এর সাথে target compare করা।

Example: Deadline-এর মধ্যে কাজ শেষ হবে কিনা দেখা।

4. Modelling

কোন action নিলে কী result হবে সেটা analyze করা।

Example: 2 জন extra employee নিলে deadline meet হবে কিনা।

5. Implementation

শেষে selected action বাস্তবে apply করা।

Exam-Ready Answer

The Management Control process consists of the following sequential steps:

1. **Data:** Collecting raw project information.
2. **Information:** Processing the data into meaningful information.
3. **Comparison:** Comparing actual performance with project objectives.
4. **Modelling:** Predicting the possible outcomes of different corrective actions.
5. **Implementation:** Carrying out the selected corrective action.

★ Shortcut

Data → Information → Comparison → Modelling → Implementation

★ Bonus Short Notes

1. ISO 12207

বাংলা অর্থ

ISO 12207 হলো একটি international standard যা software-এর **development life-cycle-এর** বিভিন্ন phase/process define করে।

Exam-Ready

ISO 12207 is an international standard that defines processes and activities for the software life-cycle, from requirements analysis through development, integration, testing, installation, and acceptance support.

2. ICCF — Four Challenges of Software

বাংলা অর্থ

Software project কেন difficult—তার চারটি কারণ:

Letter	Meaning	বাংলা
I	Invisibility	দেখা যায় না
C	Complexity	জটিলতা
C	Conformity	Existing rules/system-এর সাথে মিলতে হয়
F	Flexibility	সহজে change করা যায়

Exam-Ready

The four major challenges of software are **Invisibility, Complexity, Conformity, and Flexibility (ICCF)**.

3. Stakeholders

বাংলা অর্থ

Project-এর সাথে যাদের **interest, involvement বা stake** আছে তারা stakeholder।

যেমন:

- Customer
- User
- Developer
- Project Manager
- Management
- Implementer

Exam-Ready

Stakeholders are individuals or groups who have an interest in or are affected by a software project. They may include customers, users, developers, managers, and implementers.

4. Objective-based vs Product-based Projects

বাংলা অর্থ

Objective-based

কোনো নির্দিষ্ট **goal** achieve করাই মূল উদ্দেশ্য।

Example: Customer complaints 30% কমানো।

Product-based

একটি নির্দিষ্ট **product** তৈরি করাই মূল উদ্দেশ্য।

Example: একটি Banking Software তৈরি করা।

Exam-Ready

Objective-based projects focus on achieving a specific goal or outcome.

Product-based projects focus on developing a specific product or system.

Chapter 1 — একদম শেষ মুহূর্তের Revision

Topic	মনে রাখবে
SPM	Plan + Organize + Monitor + Control
Project	Planned + Significant
Job	Low uncertainty
Project	Medium uncertainty
Exploration	High uncertainty
ICCF	Invisibility + Complexity + Conformity + Flexibility
Management Activities	Plan, Organize, Staff, Direct, Monitor, Control, Innovate, Represent

Topic	মনে রাখবে
Project Management	Feasibility → Planning → Execution
ISO 12207	Requirements → Architecture → Code/Test → Integration → Qualification → Installation → Acceptance
Project Authority	Scope + Cost
SMART	Specific, Measurable, Achievable, Relevant, Time-constrained
Business Case	Benefits vs Costs
Management Control	Data → Information → Comparison → Modelling → Implementation
Stakeholder	Anyone with interest/stake in project
Objective-based	Achieve a goal
Product-based	Build a product

★ সবচেয়ে বেশি মুখস্থ রাখার ৫টা

1. ICCF → Software-এর 4 challenges
2. SMART → Good project objectives
3. ISO 12207 → 7 development phases
4. **Management Control** → Data → Information → Comparison → Modelling → Implementation
5. **Business Case** → Benefits should justify/exceed Costs